

SUBGROUPS OF THE SYMMETRIC GROUP: COUNTING AND ORDER STATISTICS

ABSTRACT. Let

$$s(n) = |\{H : H \leq S_n\}|.$$

The asymptotic formula is

$$\log_2 s(n) = \frac{n^2}{16} + o(n^2), \quad \text{equivalently} \quad s(n) = 2^{(1/16+o(1))n^2}.$$

The lower bound is elementary and comes from the subgroup lattice of an elementary abelian 2-subgroup of rank $\lfloor n/2 \rfloor$. The matching upper bound is the theorem of Roney-Dougal and Tracey, which proves a strong form of Pyber's 1993 conjecture. We also record what is presently available as a genuine global statistical theorem about the order of a uniformly random subgroup $H_n \leq S_n$: for every

$$\nu < \rho := \frac{1}{2} - \frac{\sqrt{3}}{4},$$

one has $\log_2 |H_n| \geq \nu n$ with probability tending to 1. This follows from their theorem that the Sylow 2-subgroup of a random subgroup has order at least $2^{\nu n}$ with high probability. Separately, inside the elementary abelian lower-bound model, a uniformly random subgroup of $C_2^{\lfloor n/2 \rfloor} \leq S_n$ has the sharper model law $2^{n/4+o(n)}$. Finally, a dihedral-block construction shows why the leading $2^{n^2/16+o(n^2)}$ enumeration alone does not determine an exact global concentration law for $|H_n|$.

1. STATEMENT OF THE ANSWER

All logarithms are base 2. Let

$$s(n) = |\text{Sub}(S_n)|$$

denote the number of subgroups of the symmetric group on n letters.

Theorem 1.1 (Asymptotic subgroup count). *As $n \rightarrow \infty$,*

$$s(n) = 2^{n^2/16+o(n^2)}.$$

Equivalently,

$$\log_2 s(n) \sim \frac{n^2}{16}.$$

The proof has two different parts. The lower bound is elementary and is proved in full below. The upper bound is a deep finite-permutation-group theorem of Roney-Dougal and Tracey. Historically, Pyber proved a weaker upper bound and conjectured the constant $1/16$; the exact asymptotic was proved by Roney-Dougal and Tracey.

For the statistical part, throughout the paper H_n denotes a uniformly random element of the finite set $\text{Sub}(S_n)$. A *section* of a finite group G means a quotient A/B , where $B \trianglelefteq A \leq G$.

Theorem 1.2 (Global order statistic currently supplied by the enumeration theory). *Let*

$$\rho = \frac{1}{2} - \frac{\sqrt{3}}{4} = 0.066987\dots$$

For every $\varepsilon > 0$,

$$\mathbb{P}(\log_2 |H_n| \geq (\rho - \varepsilon)n) \longrightarrow 1.$$

Moreover, deterministically,

$$\log_2 |H_n| \leq \log_2(n!) \leq n \log_2 n.$$

Thus a uniformly random subgroup of S_n has exponentially large order with high probability:

$$|H_n| \geq 2^{(\rho-o(1))n}$$

in the usual lower-tail sense that every smaller linear exponent holds with probability tending to 1.

Remark 1.3. Theorem 1.2 is a real statistical theorem about uniformly random subgroups of S_n , but it is not an exact concentration theorem of the form $\log_2 |H_n| \sim cn$. Such an exact global order law is a finer question than the leading enumeration $s(n) = 2^{n^2/16+o(n^2)}$. The model theorem in Section 6 and the construction in Section 7 explain this distinction.

2. GAUSSIAN BINOMIAL COEFFICIENTS

For integers $0 \leq d \leq r$, let

$$\begin{bmatrix} r \\ d \end{bmatrix}_2$$

denote the number of d -dimensional linear subspaces of $\mathbb{F}_2^r$. These numbers are the Gaussian binomial coefficients over $\mathbb{F}_2$.

Lemma 2.1 (Formula for $\begin{bmatrix} r \\ d \end{bmatrix}_2$). *For $0 \leq d \leq r$,*

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 = \frac{(2^r - 1)(2^r - 2) \cdots (2^r - 2^{d-1})}{(2^d - 1)(2^d - 2) \cdots (2^d - 2^{d-1})}.$$

Proof. Count ordered bases of d -dimensional subspaces of $\mathbb{F}_2^r$ in two ways. An ordered linearly independent d -tuple in $\mathbb{F}_2^r$ can be chosen in

$$(2^r - 1)(2^r - 2) \cdots (2^r - 2^{d-1})$$

ways: after i independent vectors have been chosen, their span has 2^i elements, so the next vector has $2^r - 2^i$ choices. On the other hand, each fixed d -dimensional subspace has

$$(2^d - 1)(2^d - 2) \cdots (2^d - 2^{d-1})$$

ordered bases, by the same argument inside $\mathbb{F}_2^d$. Dividing gives the formula. $\square$

Lemma 2.2 (Uniform Gaussian-binomial estimate). *There is an absolute constant*

$$C_2 = \prod_{j=1}^{\infty} (1 - 2^{-j})^{-1} < \infty$$

such that, for all $0 \leq d \leq r$,

$$2^{d(r-d)} \leq \begin{bmatrix} r \\ d \end{bmatrix}_2 \leq C_2 2^{d(r-d)}.$$

Proof. Starting from Lemma 2.1, factor out the powers of 2:

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 = 2^{d(r-d)} \prod_{i=0}^{d-1} \frac{1 - 2^{i-r}}{1 - 2^{i-d}}.$$

Since $r \geq d$, one has $2^{i-r} \leq 2^{i-d}$, hence

$$1 - 2^{i-r} \geq 1 - 2^{i-d}.$$

Each factor in the product is therefore at least 1, proving the lower bound.

For the upper bound, use $1 - 2^{i-r} \leq 1$. Then

$$\prod_{i=0}^{d-1} \frac{1 - 2^{i-r}}{1 - 2^{i-d}} \leq \prod_{i=0}^{d-1} (1 - 2^{i-d})^{-1} = \prod_{j=1}^d (1 - 2^{-j})^{-1} \leq \prod_{j=1}^{\infty} (1 - 2^{-j})^{-1}.$$

The infinite product converges because $\sum_{j \geq 1} 2^{-j} < \infty$. This proves the estimate. $\square$

Lemma 2.3 (Number of subspaces of $\mathbb{F}_2^r$). *Let*

$$G(r) = \sum_{d=0}^r \begin{bmatrix} r \\ d \end{bmatrix}_2.$$

Then

$$G(r) = 2^{r^2/4 + O(\log r)}.$$

In particular,

$$\log_2 G(r) = \frac{r^2}{4} + O(\log r).$$

Proof. The function $d(r-d)$ is maximized at $d = r/2$, with maximum $r^2/4$. Therefore Lemma 2.2 gives

$$G(r) \leq \sum_{d=0}^r C_2 2^{d(r-d)} \leq (r+1) C_2 2^{r^2/4}.$$

For the lower bound, take $d = \lfloor r/2 \rfloor$. Then

$$G(r) \geq \begin{bmatrix} r \\ \lfloor r/2 \rfloor \end{bmatrix}_2 \geq 2^{\lfloor r/2 \rfloor \lceil r/2 \rceil} = 2^{r^2/4 + O(1)}.$$

Combining the two estimates gives the claim. $\square$

Lemma 2.4 (Concentration of the dimension of a random subspace). *Let U_r be chosen uniformly from all linear subspaces of $\mathbb{F}_2^r$. Then*

$$\frac{\dim U_r}{r} \rightarrow \frac{1}{2}$$

in probability. Equivalently, for every fixed $\varepsilon > 0$,

$$\mathbb{P} \left(\left| \dim U_r - \frac{r}{2} \right| \geq \varepsilon r \right) \rightarrow 0.$$

More quantitatively,

$$\mathbb{P} \left(\left| \dim U_r - \frac{r}{2} \right| \geq \varepsilon r \right) \leq (r+1) C_2 2^{-\varepsilon^2 r^2 + O(1)}.$$

Proof. For every d ,

$$d(r-d) = \frac{r^2}{4} - \left(d - \frac{r}{2} \right)^2.$$

Thus, if $|d - r/2| \geq \varepsilon r$, Lemma 2.2 gives

$$\begin{bmatrix} r \\ d \end{bmatrix}_2 \leq C_2 2^{r^2/4 - \varepsilon^2 r^2}.$$

There are at most $r+1$ possible dimensions d , so the number of exceptional subspaces is at most

$$(r+1) C_2 2^{r^2/4 - \varepsilon^2 r^2}.$$

By Lemma 2.3, the total number of subspaces is at least $2^{r^2/4 + O(1)}$. Dividing the two bounds proves the quantitative estimate and hence the probabilistic convergence. $\square$

3. THE ELEMENTARY LOWER BOUND FOR $s(n)$

Let

$$m = \left\lfloor \frac{n}{2} \right\rfloor.$$

Inside S_n , define

$$E_m = \langle (1\ 2), (3\ 4), \dots, (2m-1\ 2m) \rangle.$$

Then $E_m \cong C_2^m$, because the displayed transpositions are disjoint and independent.

Proposition 3.1 (Elementary lower bound). *One has*

$$s(n) \geq 2^{n^2/16+O(n)}.$$

In fact,

$$s(n) \geq G\left(\left\lfloor \frac{n}{2} \right\rfloor\right),$$

where $G(r)$ is the number of subspaces of $\mathbb{F}_2^r$.

Proof. The subgroup E_m is naturally an m -dimensional vector space over $\mathbb{F}_2$: multiplication in E_m corresponds to addition of exponent vectors in $\mathbb{F}_2^m$. Therefore the subgroups of E_m are exactly the $\mathbb{F}_2$ -linear subspaces of $\mathbb{F}_2^m$. Distinct subspaces give distinct subgroups of S_n . Hence

$$s(n) \geq |\text{Sub}(E_m)| = G(m).$$

By Lemma 2.3,

$$G(m) = 2^{m^2/4+O(\log m)}.$$

Since $m = \lfloor n/2 \rfloor$, we have

$$\frac{m^2}{4} = \frac{n^2}{16} + O(n).$$

Thus

$$s(n) \geq 2^{n^2/16+O(n)}.$$

□

4. THE UPPER BOUND AND THE ASYMPTOTIC FORMULA

The next theorem is the non-elementary input, quoted from Roney-Dougal and Tracey [2, Theorem 1] rather than reproved.

Imported theorem 4.1 (Roney-Dougal–Tracey enumeration theorem). *There exist absolute constants $\alpha > 0$ and β such that, for all integers $n > 1$,*

$$2^{n^2/16+\alpha n \log n} \leq |\text{Sub}(S_n)| \leq 2^{n^2/16+\beta n^{3/2}}.$$

Remark 4.2. Pyber’s 1993 work gave a weaker upper bound of the form $2^{\xi n^2+o(n^2)}$, where $\xi = (\log_2 24)/6$, and conjectured the sharper constant $1/16$. Theorem 4.1 settles that conjecture. Thus the matching upper bound in this note should be attributed to Roney-Dougal and Tracey, not to Pyber alone.

Proof of Theorem 1.1. The elementary lower bound from Proposition 3.1 gives

$$s(n) \geq 2^{n^2/16+O(n)} = 2^{n^2/16+o(n^2)}.$$

The upper bound in Theorem 4.1 gives

$$s(n) \leq 2^{n^2/16+\beta n^{3/2}} = 2^{n^2/16+o(n^2)}.$$

Combining the two inequalities yields

$$s(n) = 2^{n^2/16+o(n^2)}.$$

Taking logarithms gives

$$\log_2 s(n) = \frac{n^2}{16} + o(n^2),$$

as claimed. □

5. A GLOBAL STATISTICAL THEOREM ON SUBGROUP ORDER

We now state the known global probabilistic input about uniformly random subgroups of S_n , again quoted from Roney-Dougal and Tracey [2, Theorem 6].

Imported theorem 5.1 (Roney-Dougal–Tracey random subgroup theorem). *Let H_n be uniformly random from $\text{Sub}(S_n)$. Let $\mu \in [0, 1/16)$ and let*

$$\nu \in \left[0, \frac{1}{2} - \frac{\sqrt{3}}{4}\right).$$

Then, with probability tending to 1 as $n \rightarrow \infty$, the group H_n has an elementary abelian 2-section of order at least $2^{\mu n}$, and a Sylow 2-subgroup of order at least $2^{\nu n}$.

Proof of Theorem 1.2. Let $\varepsilon > 0$. If $\varepsilon \geq \rho$, the asserted lower bound is trivial because $|H_n| \geq 1$. If $0 < \varepsilon < \rho$, put $\nu = \rho - \varepsilon$. Then $0 \leq \nu < \rho$, so Theorem 5.1 gives, with probability tending to 1, a Sylow 2-subgroup $P_n \leq H_n$ satisfying

$$|P_n| \geq 2^{\nu n} = 2^{(\rho - \varepsilon)n}.$$

Since $|P_n| \leq |H_n|$, it follows that

$$\mathbb{P}(\log_2 |H_n| \geq (\rho - \varepsilon)n) \rightarrow 1.$$

The deterministic upper bound follows from Lagrange's theorem:

$$|H_n| \leq |S_n| = n!,$$

and from the elementary inequality $n! \leq n^n$, which gives $\log_2(n!) \leq n \log_2 n$. $\square$

Remark 5.2. The elementary abelian section conclusion in Theorem 5.1 also implies $|H_n| \geq 2^{\mu n}$ with high probability for every $\mu < 1/16$, since the order of a section of a finite group divides the order of the group. The Sylow 2-subgroup assertion gives the slightly larger explicit lower-tail constant $\rho = 1/2 - \sqrt{3}/4$.

6. THE ORDER LAW INSIDE THE ELEMENTARY ABELIAN MODEL

The lower-bound construction not only gives many subgroups; it also gives a clean order statistic inside that model.

Theorem 6.1 (Model order theorem). *Let $m = \lfloor n/2 \rfloor$, and choose a subgroup $U_n \leq E_m$ uniformly at random from all subgroups of E_m . Then*

$$\log_2 |U_n| = \frac{n}{4} + o(n)$$

in probability. Equivalently,

$$|U_n| = 2^{n/4 + o(n)}$$

in probability.

Proof. Under the identification $E_m \cong \mathbb{F}_2^m$, the subgroup U_n is a uniformly random subspace of $\mathbb{F}_2^m$. Its group order is

$$|U_n| = 2^{\dim U_n},$$

so

$$\log_2 |U_n| = \dim U_n.$$

By Lemma 2.4,

$$\dim U_n = \frac{m}{2} + o(m)$$

in probability. Since $m = n/2 + O(1)$, this becomes

$$\log_2 |U_n| = \frac{n}{4} + o(n),$$

which is the claim. $\square$

7. WHY THE GLOBAL ORDER DISTRIBUTION IS A FINER QUESTION

It is tempting to infer from Theorem 6.1 that a uniformly random subgroup of S_n itself should have order $2^{n/4+o(n)}$. That inference is not justified by the leading asymptotic formula. The following construction shows why: another natural family has the same leading $2^{n^2/16}$ -scale but a different linear order scale.

Lemma 7.1 (A dihedral block group). *Let D_8 be the dihedral group of order 8, acting faithfully on four points as the symmetry group of a square. Its Frattini subgroup $\Phi(D_8)$, namely the intersection of all maximal subgroups of D_8 , has order 2, and*

$$D_8/\Phi(D_8) \cong C_2^2.$$

Proof. Write

$$D_8 = \langle r, s : r^4 = s^2 = 1, srs = r^{-1} \rangle.$$

The subgroup $\langle r^2 \rangle$ is central of order 2. The quotient $D_8/\langle r^2 \rangle$ is generated by the images of r and s , both of order 2, and these images commute because the commutator relation becomes trivial modulo $\langle r^2 \rangle$. Thus the quotient has order 4 and is isomorphic to C_2^2 .

It remains to identify $\langle r^2 \rangle$ with the Frattini subgroup. Since $|D_8| = 8$, the maximal subgroups have order 4. They are exactly

$$\langle r \rangle, \quad \langle r^2, s \rangle, \quad \langle r^2, rs \rangle.$$

Indeed, $\langle r \rangle$ is the unique cyclic subgroup of order 4, while every subgroup of order 4 not equal to $\langle r \rangle$ contains a reflection and r^2 , giving one of the two displayed Klein four subgroups according to whether the reflection is $r^{2j}s$ or $r^{2j+1}s$. Their intersection is precisely $\langle r^2 \rangle$. Hence

$$\Phi(D_8) = \langle r \rangle \cap \langle r^2, s \rangle \cap \langle r^2, rs \rangle = \langle r^2 \rangle.$$

The quotient by $\Phi(D_8)$ is therefore C_2^2 , as proved above. $\square$

Proposition 7.2 (Many subgroups at order scale $2^{n/2}$). *Suppose $n = 4k$. Then S_n contains a family $\mathcal{F}_n$ of subgroups with*

$$|\mathcal{F}_n| = 2^{n^2/16+O(\log n)}$$

such that all but an exponentially small proportion of the members of $\mathcal{F}_n$ satisfy

$$\log_2 |H| = \frac{n}{2} + o(n).$$

Proof. Partition $\{1, \dots, n\}$ into k fixed blocks of size 4. On each block let a copy of D_8 act as in Lemma 7.1, and set

$$P_k = D_8^k \leq S_{4k}.$$

Let

$$N_k = \Phi(D_8)^k \trianglelefteq P_k.$$

Then

$$N_k \cong C_2^k \quad \text{and} \quad P_k/N_k \cong C_2^{2k}.$$

Let

$$\pi : P_k \rightarrow P_k/N_k$$

be the quotient map. For every subspace $U \leq P_k/N_k$, the inverse image

$$H_U = \pi^{-1}(U)$$

is a subgroup of P_k , hence a subgroup of S_{4k} . Distinct subspaces U give distinct subgroups H_U , since $\pi(H_U) = U$. Therefore the number of such subgroups is exactly the number of subspaces of $\mathbb{F}_2^{2k}$, namely

$$G(2k) = 2^{(2k)^2/4+O(\log k)} = 2^{k^2+O(\log k)}.$$

Since $n = 4k$, this is

$$2^{n^2/16+O(\log n)}.$$

Moreover, if $\dim U = d$, then

$$|H_U| = |N_k| |U| = 2^k 2^d = 2^{k+d}.$$

By Lemma 2.4, a uniformly random subspace $U \leq \mathbb{F}_2^{2k}$ has

$$d = k + o(k)$$

in probability. Hence, for all but an exponentially small proportion of the subspaces U ,

$$\log_2 |H_U| = k + d = 2k + o(k) = \frac{n}{2} + o(n).$$

This proves the claim. $\square$

Corollary 7.3 (The leading formula alone does not give an exact global order law). *The asymptotic formula*

$$s(n) = 2^{n^2/16+o(n^2)}$$

does not by itself imply that a uniformly random subgroup of S_n has order $2^{n/4+o(n)}$. Indeed, for $4 \mid n$, there is a family of

$$2^{n^2/16+O(\log n)}$$

subgroups of order scale $2^{n/2+o(n)}$, as shown in Proposition 7.2.

Proof. Theorem 6.1 proves the order scale $2^{n/4+o(n)}$ only for subgroups chosen inside a fixed elementary abelian subgroup of rank $\lfloor n/2 \rfloor$. Proposition 7.2 constructs, on the same leading $2^{n^2/16}$ logarithmic scale, subgroups whose order is instead $2^{n/2+o(n)}$. Therefore the leading exponential count cannot distinguish the global probability weights of these different families. A sharper theorem for a uniformly random subgroup of S_n requires finer enumeration than the main logarithmic asymptotic. $\square$

8. A USEFUL RANK CHECK FOR ELEMENTARY ABELIAN 2-SUBGROUPS

The following elementary observation explains why the rank $\lfloor n/2 \rfloor$ appearing in the lower bound is the correct maximum possible rank for elementary abelian 2-subgroups of S_n .

Lemma 8.1. *If $A \leq S_n$ is elementary abelian of exponent 2, say $A \cong C_2^r$, then*

$$r \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Let A act on $\{1, \dots, n\}$, and decompose the set into A -orbits

$$\Omega_1, \dots, \Omega_t.$$

For each orbit Ω_i , the image of A in $\text{Sym}(\Omega_i)$ is a transitive abelian permutation group. A transitive abelian permutation group acts regularly after factoring out its kernel: if an element fixes one point of a transitive orbit, then, because the group is abelian, it fixes every point of that orbit. Hence the faithful image on Ω_i is an elementary abelian 2-group of order $|\Omega_i|$. Thus $|\Omega_i| = 2^{a_i}$ for some $a_i \geq 0$, and the rank of the image on Ω_i is a_i .

The diagonal action of A on all its orbits is faithful, so the natural homomorphism from A to the product of its images on the orbits is injective. Therefore

$$r \leq \sum_{i=1}^t a_i.$$

For each orbit, $a_i \leq 2^{a_i}/2$ when $a_i \geq 1$, and the inequality is also true for $a_i = 0$. Hence

$$r \leq \sum_{i=1}^t a_i \leq \frac{1}{2} \sum_{i=1}^t 2^{a_i} = \frac{n}{2}.$$

Since r is an integer, $r \leq \lfloor n/2 \rfloor$. $\square$

9. VERIFICATION OF THE ANSWER

We finish by isolating exactly what has been proved and where each input enters.

- (1) Lemmas 2.1–2.4 are fully elementary linear algebra over $\mathbb{F}_2$. They prove both the count and dimension concentration for subspaces of $\mathbb{F}_2^r$.
- (2) Proposition 3.1 injects the subgroup lattice of $C_2^{\lfloor n/2 \rfloor}$ into $\text{Sub}(S_n)$. This proves

$$s(n) \geq 2^{n^2/16+O(n)}.$$

- (3) The non-elementary upper bound is Theorem 4.1, due to Roney-Dougal and Tracey. Combining it with the elementary lower bound gives

$$s(n) = 2^{n^2/16+o(n^2)}.$$

This is the requested asymptotic formula.

- (4) The genuine global statistical theorem on order is Theorem 1.2: if H_n is uniformly random in $\text{Sub}(S_n)$, then for every $\varepsilon > 0$,

$$\mathbb{P} \left(\log_2 |H_n| \geq \left(\frac{1}{2} - \frac{\sqrt{3}}{4} - \varepsilon \right) n \right) \rightarrow 1.$$

This is obtained from the Roney-Dougal–Tracey random subgroup theorem.

- (5) Theorem 6.1 is a fully proved sharper statistical theorem for the elementary abelian lower-bound model. It says that a random subgroup of $C_2^{\lfloor n/2 \rfloor} \leq S_n$ has order $2^{n/4+o(n)}$.
- (6) Proposition 7.2 and Corollary 7.3 explain why the leading logarithmic enumeration alone does not force a unique exact global order scale.

Thus the original question is answered as follows: the number of subgroups of S_n is $2^{n^2/16+o(n^2)}$; there is a global high-probability theorem proving that a uniformly random subgroup has exponentially large order, and there is an exact order law in the elementary abelian model that supplies the elementary lower bound. A sharper global concentration theorem for $\log_2 |H_n|$, if desired, requires finer information than the leading asymptotic count.

REFERENCES

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